CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Geometric Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The general term

1. 1701 is the nth term of the G.P. $7, 21, 63, \ldots$ Find (a) n (b) the sum of these n terms. ICSE 2025, 3 marks

SOLUTION

(a) $a = 7$, $r = 3$. $7 \times 3^{n-1} = 1701$, so $3^{n-1} = 243 = 3^5$ and $n = 6$.

(b) $S_6 = \dfrac{a(r^6 - 1)}{r - 1} = \dfrac{7(729 - 1)}{2} = 7 \times 364$

Answer: (a) $n = 6$ (b) sum $= 2548$

2. The 4th, 6th and last terms of a G.P. are 10, 40 and 640. The common ratio is positive. Find the first term, the common ratio and the number of terms. ICSE 2020, 4 marks

SOLUTION

$ar^3 = 10$ and $ar^5 = 40$. Dividing: $r^2 = 4$, so $r = 2$.

$a = \dfrac{10}{8} = \dfrac{5}{4}$

Last term: $\dfrac{5}{4} \times 2^{n-1} = 640$, so $2^{n-1} = 512 = 2^9$ and $n = 10$.

Answer: $a = \dfrac{5}{4}$, $r = 2$, $n = 10$

3. The 4th term of a G.P. is 16 and the 7th term is 128. Find the first term and the common ratio. ICSE 2018, 3 marks

SOLUTION

$ar^3 = 16$ and $ar^6 = 128$. Dividing: $r^3 = 8$, so $r = 2$.

$a = \dfrac{16}{8} = 2$

Answer: first term 2, common ratio 2

MathSetu

2 Sum of n terms

4. $15, 30, 60, 120, \ldots$ is a G.P. (a) Write its nth term. (b) How many terms add up to 945? ICSE 2024, 4 marks

SOLUTION

(a) $a = 15$ and $r = 2$, so $T_n = 15 \times 2^{n-1}$.

(b) $S_n = \dfrac{a(r^n - 1)}{r - 1} = 15(2^n - 1) = 945$

$2^n - 1 = 63$, so $2^n = 64 = 2^6$.

Answer: (a) $T_n = 15 \times 2^{n-1}$ (b) 6 terms

5. A G.P. has first term 3, last term 96 and common ratio 2. Find (a) the number of terms (b) their sum. ICSE 2019, 3 marks

SOLUTION

(a) $3 \times 2^{n-1} = 96$, so $2^{n-1} = 32 = 2^5$ and $n = 6$.

(b) $S_6 = \dfrac{3(2^6 - 1)}{2 - 1} = 3 \times 63$

Answer: (a) 6 terms (b) 189

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.